A Neutrino Mixing Sum Rule from Octahedral Flavor Structure: 2 $\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$

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Abstract

Discrete flavor symmetries yield characteristic correlations among the PMNS mixing angles, with the trimaximal TM1 and TM2 sum rules serving as benchmark targets for the current generation of reactor experiments. We present a distinct pattern arising from octahedral (cubic-group) flavor structure: tri-bimaximal mixing at zeroth order, corrected at $O(\lambda^2)$ by a perturbation whose scale $\lambda = 1/(\pi\sqrt{2}) = 0.22508$ is fixed by the Brillouin-zone geometry of an underlying lattice flavor construction [18] and coincides with the observed Cabibbo angle to 0.04%. The resulting angles are $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.30798$, $\sin^2\theta_{13} = 4/(18\pi^2) = 0.02252$, and $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.55066$, with no parameter fitted to any mixing angle; the system is closed by one Wilson-coefficient relation (Condition 2), stated as an explicit condition whose derivation from the underlying construction remains open. The pattern agrees with the post-JUNO global fit $\sin^2\theta_{12} = 0.3085 \pm 0.0073$ at 0.07σ — a retrodiction, the derivation postdating JUNO's first results (November 2025), with no claim of temporal priority — and with NuFIT 6.0 within 1.1σ on the remaining angles. The forward content is twofold: the exact sum rule 2 $\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ (currently satisfied at 0.6σ), which discriminates this pattern from the TM1 and TM2 patterns at JUNO precision, and the exposure of the exact value $\sin^2\theta_{12} = 0.30798$ to JUNO's design-lifetime precision of ± 0.0014 — a ~4σ discriminator if the current central value holds.

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1. Introduction

1.1 The result

Tri-bimaximal mixing [1] was the leading description of the lepton mixing pattern until Daya Bay established a nonzero θ_{13} [2]. The minimal modified patterns that followed — TM1 and TM2, each preserving one column of the TBM matrix — remain the benchmark correlations against which reactor-era precision data are analyzed [3, 4, 5, 6], and the first JUNO measurement has already sharpened this comparison, excluding TM2 at more than 3.5σ while leaving TM1 viable [9, 10]. Patterns in this class are typically derived from a discrete flavor symmetry postulated for the lepton sector. In this paper we present a TBM-deviation pattern of a different origin, with different — and sharper — testable consequences.

The first JUNO measurement of reactor neutrino oscillations [7] reports

$\mathbf{\sin^2\theta_{12} = 0.3092 \pm 0.0087}$

from 59.1 days of data. The post-JUNO global fit of Capozzi *et al.* [8] tightens this to

$\mathbf{\sin^2\theta_{12} = 0.3085 \pm 0.0073}$.

We show that taking the cubic point group O — the rotation symmetry group of three-dimensional space — as the underlying flavor structure of the lepton sector yields the prediction

$\mathbf{\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.30798}$

matching the post-JUNO global fit at $\mathbf{0.07\sigma}$ with no parameter fit to $\sin^2\theta_{12}$. The same construction predicts $\sin^2\theta_{13}$ and $\sin^2\theta_{23}$ within 1.1σ of the NuFIT 6.0 best fits [14] and forces the exact sum rule

$\mathbf{2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6}$

which distinguishes the present prediction from the TM1, TM2, and TM3 column-preservation patterns at JUNO precision. Both inputs to $\sin^2\theta_{12}$ are independently motivated: the zeroth-order value 1/3 is the A_4 TBM result, and the perturbation scale $\lambda^2 = 1/(2\pi^2)$ is the squared Cabibbo angle, fixed independently by the cubic Brillouin geometry (matching the observed λ to 0.04%).

1.2 Where the prediction comes from

The structure is constrained by two ingredients. First, the perturbation scale $\lambda^2 = 1/(2\pi^2)$ is not a fitted parameter: it is the squared Cabibbo angle, fixed by a single cubic Brillouin-zone distance (Sec. 2.2). Numerically $\lambda = 1/(\pi\sqrt{2})$ = 0.22508, agreeing with the observed Cabibbo value 0.22500 ± 0.00067 [13] to 0.04%. Second, the deviations from tribimaximal (TBM) mixing at $O(\lambda^2)$ are controlled by a discrete μ-τ parity (the U-parity of $S_4 \supset A_4$) together with one Wilson-coefficient relation between the U-even and U-odd components of the second-order perturbation — *Condition 2*, with the specific value $b_{23} = (4/3)(b_{12}+b_{13})$ admitting the structural reading $4/3 = 2A^2 = 4(d-1)/(2d)$ for $d = 3$ (Sec. 3.4). Condition 2 fixes the coefficient of the $1/(4\pi^2)$ correction; without further inputs, the angle predictions are

- $\mathbf{\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.30798}$
- $\mathbf{\sin^2\theta_{13} = 4/(18\pi^2) = 0.02252}$
- $\mathbf{\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.55066}$

The 0.07σ match with the post-JUNO global fit therefore tests Condition 2 — a sharply specified structural relation — rather than fitted parameters.

1.3 The empirical context

The cubic-group structure used here is not a postulated flavor symmetry. It is the rotation symmetry group of three-dimensional space, realized in a lattice-based framework [18] in which the simple cubic Bravais lattice in d=3 is shown to be the unique three-dimensional Bravais lattice consistent with the SU(3) × SU(2) × U(1) gauge structure and the observed Cabibbo angle (Theorem 7b). The same cubic structure that produces the $1/(4\pi^2)$ correction here also produces a number of related quantitative predictions across the flavor sector, all empirically tested:

Observable	Cubic-group prediction	Observed	Match
Cabibbo angle λ	$1/(\pi\sqrt{2}) = 0.22508$	0.22500 ± 0.00067 [13]	0.04%
Koide angle θ_0 (charged leptons)	$C_2/d^2 = 2/9$	0.222204 ± 0.000010 [13]	0.02%
Solar mixing $\sin^2\theta_{12}$ (this work)	$1/3 - 1/(4\pi^2) = 0.30798$	0.3085 ± 0.0073 [8]	0.07σ
PMNS sum rule	$2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$	1.178 ± 0.020	0.6σ

The first two entries are derived in [18, §§7.1–7.2]; the Cabibbo geometry and the chirality-forced $1/|q|$ power are unconditional structural (the chirality structure of the taste-changing vertex is established in the staggered fermion literature [18, §7.1, citing Mason et al.]), while the identification of the mixing-matrix element with the unit-normalized propagator is a stated structural hypothesis whose resolving computation is specified in [18, §7.1], and the empirical match to 0.04% supports this status. The Koide angle is structural with its per-axis normalization forced by cubic symmetry; the overall amplitude prefactor and the representation carrier of the mass-splitting operator are stated inputs whose resolving computations are specified in [18, §7.2, §4.7]. The third and fourth entries are the present work, classified at the layered level (Layer 1 form combined with the Layer 2(b) Condition 2 coefficient) per the operational definitions in §3.4; see Sec. 6.3 for full discussion. This cumulative empirical track record is what distinguishes the cubic-group structure from postulated flavor symmetries: the same parameter $\lambda^2 = 1/(2\pi^2)$ and the same projection factor $A^2 = 2/3$ enter multiple observables at sub-percent precision, which is testable structure rather than symmetry assumption.

1.4 Position in the broader literature

The lattice construction [18] from which the cubic-group structure is taken belongs to a recent line of work in which observers play a structural role in the emergence of quantum theory [19, 20, 21, 22]; it differs from those constructions in being deterministic at the base level and in producing the cubic flavor structure used here as a consequence rather than an input. None of that context is load-bearing for the present analysis: the flavor pattern of Secs. 2–5 stands as a self-contained group-theoretic construction, and its comparison with data (Secs. 4–7) is independent of how the underlying lattice is interpreted.

The remainder of this paper develops the technical content. Section 2 sets up the cubic-group structure and derives the perturbation scale. Section 3 computes the deviations from TBM at $O(\lambda^2)$ and derives the sum rule. Section 4 presents the predictions and compares them with JUNO, the post-JUNO global fit, and competing TM1/TM2 patterns. Section 5 covers the remaining angles. Section 6 addresses robustness. Section 7 discusses falsifiability under JUNO's projected long-term precision. Section 8 concludes.

1.5 Provenance

This derivation was carried out in 2026, after the JUNO first results [7] and the post-JUNO global fit [8] appeared. The value was not fitted to the measurement — it is fixed by the cubic-group structure with no adjustable freedom — but the match is retrodictive, and this paper makes no claim of temporal priority over the data. Its falsifiable forward content is threefold: (i) the exact value 0.30798 against JUNO's design-lifetime precision of ±0.0014 (§8 of the design report; a ~4σ discriminator if the central value holds); (ii) the exact sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ against the TM1 and TM2 patterns analyzed in the post-JUNO literature [9, 10]; and (iii) the θ_{13} and θ_{23} values as global precision sharpens. This places the paper within the post-JUNO structural-interpretation literature [9, 10], with its provenance stated explicitly.

2. Cubic-group flavor structure

2.1 The cubic group O and its A_4 subgroup

The cubic point group O is the rotation symmetry group of three-dimensional space, the unique finite point group with a six-element generator set $\{\pm\hat{e}_1, \pm\hat{e}_2, \pm\hat{e}_3\}$ — six minimal-length translations corresponding to forward and backward motion along three orthogonal axes. In the lattice realization of the OI framework, this six-element set arises from coupling-degree minimization ($K = 2d = 6$ for $d = 3$), and the multiplicities (3, 2, 1) under the action of O on these six directions match the structure of one Standard Model fermion generation (quark triplet, lepton doublet, singlet). Three SM generations are placed on the T_1 representation of the cubic group O acting on the three Cartesian axes — a forced consequence of [18, §4.6 Theorem 7] (cubic decomposition $6 = T_1 \oplus E \oplus A_1$), [18, §4.6 Theorem 7b] (Bravais-lattice uniqueness — SC is the unique 3D Bravais lattice compatible with the SM gauge group at the commutant level and with the observed Cabibbo angle), [18, §4.7 Theorem 10] (cubic symmetry forces three degenerate triplet sectors), [18, §4.7.1.2] (link-carrier construction placing matter on links with cross-charged representations admissible), and [18, Theorem 15] (anomaly cancellation fixing hypercharges uniquely). The detailed argument from coupling-degree minimization through the (3, 2, 1) decomposition is given in [18, §§4.5–4.7].

The full octahedral rotation group O has order 24 and is isomorphic to S_4. Its irreducible representations decompose as

$$\mathbf{irreps(O) = A_1 \oplus A_2 \oplus E \oplus T_1 \oplus T_2}$$

of dimensions 1, 1, 2, 3, 3. The subgroup $A_4 \subset O$ keeps $\{A_1, A_2, E, T_1\}$ but identifies A_2 with A_1, giving the standard A_4 content A ⊕ A' ⊕ A'' ⊕ T of dimensions 1, 1, 1, 3.

We take T_1 as the generation triplet. Realize it by the three Cartesian unit vectors $\{e_1, e_2, e_3\}$ acted on by the 24 rotations of O. The democratic vector

$$\mathbf{\hat{h} = (1/\sqrt{3})(e_1 + e_2 + e_3)}$$

is the only direction fixed by the $\mathbb{Z}_3$ cyclic subgroup. The plane perpendicular to $\hat{h}$ carries the doublet E. This is the geometric origin of TBM: the second column of U_TBM is exactly $\hat{h}$, and the first and third columns lie in the plane perpendicular to it, oriented by μ-τ exchange.

2.1.1 Why the cubic group is forced

The cubic group is not postulated here as a flavor symmetry of the lepton sector in the manner of A_4, S_4, or $\Delta(27)$ in the discrete-flavor-symmetry literature. It is the unique three-dimensional Bravais-lattice point-group symmetry compatible with the Standard Model gauge group SU(3) × SU(2) × U(1) under the commutant construction of [18, §4.6 Theorem 7b]. The argument is in two parts.

(a) Non-cubic Bravais lattices are excluded by crystallographic representation theory. Among the 32 crystallographic point groups, three-dimensional irreducible representations appear *only* in the cubic crystal system (point groups T, T_h, T_d, O, O_h). Point groups of the hexagonal, trigonal, tetragonal, orthorhombic, monoclinic, and triclinic systems act on $\mathbb{R}^3$ reducibly as (axial direction) + (perpendicular plane) and therefore have irreps only of dimensions 1 and 2 — no 3-dimensional irrep. The commutant construction producing an SU(3) gauge factor requires a 3-dimensional irreducible block in the coupling matrix M, which non-cubic point groups cannot supply. This eliminates 11 of the 14 three-dimensional Bravais lattices at the group-theoretic level, independent of any numerical prediction.

(b) BCC and FCC are excluded within the cubic system. Computing the explicit O-decomposition of the nearest-neighbor link representation on each of the three cubic Bravais lattices [18, §4.6 Theorem 7b]:

Lattice	NN	O-decomposition of NN rep	Gauge group (commutant)
SC	6	$T_1 \oplus E \oplus A_1$	SU(3) × SU(2) × U(1) ✓
BCC	8	$A_1 \oplus A_2 \oplus T_1 \oplus T_2$	$\mathrm{SU}(3)^2 \times \mathrm{U}(1)^2$
FCC	12	$A_1 \oplus E \oplus T_1 \oplus 2T_2$	$\mathrm{SU}(3)^3 \times \mathrm{SU}(2) \times \mathrm{U}(1)$

SC is uniquely compatible with the observed Standard Model gauge structure. BCC gives two non-abelian SU(3) factors with no SU(2); FCC gives three SU(3) factors and an extra SU(3) factor beyond observation. Neither matches at the group-theoretic level. The same prescription applied within each lattice's natural triplet sector then gives the Cabibbo angle: SC gives $\lambda = 1/(\pi\sqrt{2}) = 0.2251$ (0.04% match), BCC gives $\lambda = 0.1299$ (42% discrepant), FCC gives $\lambda = 0.1592$ (29% discrepant). The SC quantitative match is at the level of structural relations and is consistent with the group-theoretic selection.

The cubic group O is therefore derived rather than chosen: it is the unique Bravais-lattice symmetry compatible with both the Standard Model gauge group and the observed Cabibbo angle among all 14 three-dimensional Bravais lattices. The A_4 subgroup of O used in the TBM-style analysis of §2.1 and §3 is the natural subgroup acting on the three-generation T_1 representation, with the spatial rotation group fixed by the gauge-structure derivation of [18, Theorem 7b] rather than postulated as a flavor symmetry.

2.2 The Cabibbo scale from cubic Brillouin geometry

We identify the perturbation parameter λ with the Cabibbo angle. In a lattice realization of the cubic-group flavor structure, the three generations occupy the three T_1 corners of the cubic Brillouin zone at momenta X_j = $\pi \cdot$ e_j (j = 1, 2, 3). The inter-generation distance in momentum space is

|X_i − X_j| = $\pi\sqrt{2}$ (i ≠ j)

a fixed geometric constant of the cubic lattice. The leading-order inter-generation mixing matrix element is the continuum fermion propagator at this momentum, |M_ij| = |S(X_j − X_i)| = 1/|X_j − X_i|. The 1/|q| scaling (rather than $1/|q|^2$) is structural: in the massless limit, the chirality-preserving component of the fermion propagator at momentum q reduces to $S(q) = \gamma \cdot q/q^2$, and the taste-changing vertex's Dirac trace contracts this to a magnitude of order 1/|q| rather than the full propagator's $1/|q|^2$. The full derivation, including the chirality verification of the taste-changing vertex's spinor structure, is given in [18, §7.1]. Identifying the matrix element with the Cabibbo angle gives

$\lambda = 1/(\pi\sqrt{2}) = 0.22508$

matching the observed value $\lambda\_obs = 0.22500 \pm 0.00067$ [13] to 0.04%. Squaring,

$$\boldsymbol{\lambda^2 = 1/(2\pi^2) = 0.05066}$$

sets the overall scale of $O(\lambda^2)$ corrections. The full derivation of $\lambda = 1/(\pi\surd 2)$ from the lattice fermion propagator and the chirality-preservation argument is given in [18, §7.1]. One element of the chain is graded separately there: the identification of the mixing-matrix element with the unit-normalized propagator (coefficient exactly one) is a stated hypothesis rather than a derived consequence, with the resolving computation — the taste-changing two-point function with explicit external-leg normalization — specified in [18, §7.1]; the geometry $(\pi\surd 2)$ and the $1/|q|$ power are forced. The empirical match to 0.04% confirms the identification at the level of structural relations rather than fitted parameters, but does not derive its unit coefficient.

A second geometric quantity will appear. The angle between any generation axis e_i and the democratic direction $\hat{h}$ satisfies $e\_i \cdot \hat{h} = 1/\surd 3$, so $\cos^2(\angle(e\_i, \hat{h})) = 1/3$ and $\sin^2(\angle(e\_i, \hat{h})) = 2/3$. We identify the Wolfenstein parameter A with this quantity:

$$\mathbf{A = \surd(2/3) = 0.8165}$$

agreeing with the PDG value $A = 0.826 \pm 0.012$ at 0.23σ. The combination $A^2 = 2/3$ enters the inter-generation projection geometry, and $A^4 = 4/9$ appears directly in the deviation Δ_{13} below.

2.3 The PMNS matrix in the Altarelli-Feruglio basis

We work in the Altarelli-Feruglio (AF) basis [12]. In this basis the neutrino sector realizes TBM exactly (after diagonalization of the seesaw matrix), and all deviations from TBM come from the charged-lepton mass matrix. The PMNS matrix is

$$\mathbf{U\_PMNS = V\_\ell\dagger \cdot U\_TBM}$$

where V_ℓ is the charged-lepton rotation that diagonalizes $M\_\ell\, M\_\ell\dagger$, and

```
              ( √(2/3)   √(1/3)    0      )
U_TBM    =    | -√(1/6)   √(1/3)  -√(1/2)  |
              ( -√(1/6)   √(1/3)   √(1/2)  )
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Expanding V_ℓ in λ,

$$\mathbf{V\_\ell = \exp(\lambda A_1 + \lambda^2 A_2 + O(\lambda^3))}$$

with A_1, A_2 real antisymmetric 3×3 matrices. The remaining task is to determine A_1 and A_2 from the structural ingredients of the cubic geometry.

3. U-parity grading and the sum rule

3.1 The U-parity generator

In the AF basis, the μ-τ exchange generator $U \in S_4$ acts on the generation indices by exchanging the second and third:

```
           ( 1   0   0 )
U    =     | 0   0   1 |
           ( 0   1   0 )
```

Conjugation by U defines a $\mathbb{Z}_2$ grading on antisymmetric matrices: A is **U-even** if $UAU^{-1} = +A$, **U-odd** if $UAU^{-1} = -A$. With (E_{ij})*{kl}* = δ{ik}δ_{jl} − δ_{il}δ_{jk}, the decomposition is:

- **U-even**: $E_{12} + E_{13}$
- **U-odd**: $E_{12} - E_{13}$, E_{23}

The unique U-odd combination antisymmetric in (μ, τ) is E_{23}; the unique U-odd combination relating the first generation to the second/third is $E_{12} - E_{13}$.

3.2 Determination of A_1

A_1 is fixed by three structural conditions plus the projection geometry of §2.2.

The data require $\sin\theta_{13} = O(\lambda)$ (since $\theta_{13} \neq 0$), forcing A_1 to contribute to the (1,3) element of V_ℓ† U_TBM. The observation that both $\Delta_{12} \equiv \sin^2\theta_{12} - 1/3$ and $\Delta_{23} \equiv \sin^2\theta_{23} - 1/2$ are $O(\lambda^2)$ (not $O(\lambda)$) further constrains A_1's structure. The remaining input is geometric: each end of an inter-generation mixing vertex projects onto the plane perpendicular to ĥ, contributing a factor sin(∠(e_i, ĥ)) = A from the off-democratic projection (Sec. 2.2). For a single-vertex inter-generation transition, the total projection factor is A^2 (one factor per vertex end). Identifying this projection geometry with the leading $O(\lambda)$ contribution to U_e3 gives the structural identity

$$\mathbf{\sin\theta_{13} = A^2\lambda}$$

(at $O(\lambda)$; A_2 contributes only at $O(\lambda^2)$ and cancels from $|U\_e3|^2$ as shown in §3.3 below). This is the structural input — not a derived consequence — that fixes A_1's overall coefficient up to sign.

The result is

$$\mathbf{A_1 = (\surd 2/3)(E_{12} - E_{13})}$$

A_1 is purely U-odd: $UA_1U^{-1} = -A_1$. Geometrically, it is a rotation by angle $A^2 = 2/3$ around the axis $(e_2 + e_3)/\surd 2$, which reproduces $\sin\theta_{13} = A^2\lambda$ as required.

3.3 Deviations to $O(\lambda^2)$

Expanding V_ℓ to second order,

$$\mathbf{V\_\ell = I + \lambda A_1 + \lambda^2(A_2 + ½A_1{}^2) + O(\lambda^3)}$$

and writing the most general antisymmetric A_2 as

$$\mathbf{A_2 = b_{12}E_{12} + b_{13}E_{13} + b_{23}E_{23}}$$

we compute |U_PMNS,ij|^2 to O(λ^2). The PMNS angles satisfy $\sin^2\theta_{13}$ = |U_e3|^2, $\sin^2\theta_{12}$ $\cos^2\theta_{13}$ = |U_e2|^2, and $\sin^2\theta_{23}$ $\cos^2\theta_{13}$ = |U_μ3|^2. Direct computation (taking V_ℓ^T = exp(-(λA_1+$\lambda^2 A_2$)) acting on the columns of U_TBM, using the explicit form $A_1{}^2$ = diag(-4/9, -2/9, -2/9) + (2/9)($e_2 \otimes e_3$ + $e_3 \otimes e_2$) for A_1 = (√2/3)($E_{12} - E_{13}$)) gives the deviations from TBM:

- $\mathbf{\Delta_{13} = (4/9)\lambda^2 = A^4\lambda^2}$
- $\mathbf{\Delta_{12} = -(2/3)(b_{12} + b_{13})\ \lambda^2}$
- $\mathbf{\Delta_{23} = b_{23}\ \lambda^2}$

Three observations follow.

(i) $\mathbf{\Delta_{13}}$ is fully determined by $\mathbf{A_1}$ alone. The A_2 parameters cancel from |U_e3|^2 at O(λ^2), so $\sin^2\theta_{13}$ = (4/9)λ^2 is fixed without reference to A_2. Equivalently, the structural identity $\sin\theta_{13}$ = $A^2\lambda$ is exact at O(λ) and receives no O(λ^2) correction.

(ii) The combination $\mathbf{b_{12} - b_{13}}$ does not enter any of the three angles. It is naturally identified with the CP-violating Dirac phase δ_CP, since it is the only remaining free parameter at O(λ^2) that does not enter the angles. The present analysis leaves δ_CP undetermined.

(iii) $\mathbf{\Delta_{23}}$ depends only on $\mathbf{b_{23}}$, not on the U-even combination. The explicit calculation gives |U_μ3|^2 = ½ + λ^2(b_{23} − 2/9) before the $\cos^2\theta_{13}$ dressing; after dressing, $\sin^2\theta_{23}$ = ½ + λ^2 b_{23} exactly. The U-even combination (b_{12}+b_{13}) does not contribute to Δ_{23} at O(λ^2): the contributions that would enter through the (V_ℓ^T)$_2\nu$ matrix elements all arise at O(λ^3) or higher in the calculation of |U_μ3|^2.

3.4 Sum rule and absolute normalization

The deviations depend on two scalar combinations of A_2: (b_{12} + b_{13}) controlling Δ_{12}, and b_{23} controlling Δ_{23}. We fix the normalization with two conditions.

Condition 1 (natural normalization). Take b_{23} = 1. This sets A_2 to enter the perturbation series at unit strength relative to λ^2. It is a labeling convention rather than a physical input — fixing the normalization of the O(λ^2) expansion before b_{12} + b_{13} is determined.

Condition 2 (cubic-group structural relation). Take

$$\mathbf{b_{23} = (4/3)(b_{12} + b_{13})}$$

which with Condition 1 gives b_{12} + b_{13} = 3/4. Unlike Condition 1, this is a substantive physical input: it relates the dominant U-odd coefficient of A_2 to the U-even combination through a fixed numerical factor 4/3.

Equivalence with TBM sum rule preservation. Combining the explicit results $\Delta_{12} = -(2/3)(b_{12}+b_{13})\lambda^2$ and $\Delta_{23} = b_{23}\lambda^2$ from §3.3:

$$\mathbf{2\Delta_{12} + \Delta_{23} = (b_{23} - (4/3)(b_{12}+b_{13}))\ \lambda^2}$$

Hence Cond 2 is *equivalent* to the statement that the TBM sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ — which holds at zeroth order from cubic geometry, since $2(1/d) + 1/(d-1) = (3d-2)/(d(d-1)) = 7/6$ for $d = 3$ — continues to hold at $O(\lambda^2)$. The Wilson-coefficient relation and the angle-level sum rule are the same content under different normalizations.

Structural protection of the sum rule by A_1 alone (Layer 1). With $A_2 = 0$, direct calculation gives $2|U\_e2|^2 + |U\_\mu 3|^2 = 7/6 + \lambda^2[-14/27]$ + ($\cos^2\theta_{13}$ corrections). Dividing by $\cos^2\theta_{13} = 1 - (4/9)\lambda^2$ (which itself follows from $\sin^2\theta_{13} = A^4\lambda^2$, a Layer 1 quantity) precisely cancels the $-14/27$ offset:

$(2\sin^2\theta_{12} + \sin^2\theta_{23})|\_\{A_2{=}0\} = 7/6 + O(\lambda^4)$

— that is, the sum rule is preserved by A_1 alone at all orders up to λ^2. This cancellation is non-trivial and specific to A_1's Layer 1-forced form: the magnitude ($\sqrt{}2/3$) is fixed by $\sin\theta_{13} = A^2\lambda$ (§3.2), and the U-odd character is fixed by the requirement Δ_{12}, $\Delta_{23} = O(\lambda^2)$ rather than $O(\lambda)$. With those two constraints, the cancellation between the $-14/27$ contribution from $A_1{}^2$-induced shifts and the $+14/27$ contribution from the $\cos^2\theta_{13}$ dressing is automatic. The TBM sum rule is therefore *structurally protected* against A_1's perturbation by cubic geometry alone — no substrate input required for this component.

Cond 2 reduces to the question of whether A_2 preserves the sum rule it was protected for under A_1, or breaks it. The answer depends on the substrate's specific b coefficients.

The factor 4/3. Cond 2's specific numerical coefficient admits the structural reading

$$\mathbf{4/3 = 2A^2 = 2\ \sin^2(\angle(e\_i,\ \hat{h}))}$$

where $A = \sqrt{}(2/3)$ is the off-democratic projection of any generation axis (Sec. 2.2). The identity $A^2 = (d-1)/d = C_2(T_1)/d$ for $d = 3$ makes this a representation-theoretic quantity rather than a numerical coincidence: it is the squared off-democratic projection for the vector representation of any rotation group. The factor 4/3 in Condition 2 admits a *partner-count × per-partner projection* reading: the U-even combination $E_{12}+E_{13}$ couples generation e_1 to its two partners $\{e_2,\ e_3\}$, with the per-partner projection nominally A^2. However, the load-bearing claim of this argument is the *operator alignment* — that the substrate's second-order Yukawa structure has a specific T_2 direction matching the partner-count × projection picture. Operator counting on $T_1 \otimes T_1 \to T_2$ in the supporting framework [18] shows that this T_2 direction is determined by the substrate's specific Yukawa structure rather than forced by representation theory: the natural Higgs-democratic spurion does not produce Cond 2.

Layered classification of predictions. Before applying this terminology to Condition 2, we state operational definitions that classify predictions by what is required to derive them. The framework's predictions stratify into four layers:

- **Layer 0 — Gauge and discrete structure.** Properties of the equivalence class of substratum bijections that follow from the framework's structural conditions plus cubic-group representation theory. Hold for *every* bijection in the class. Examples: the SM gauge group with multiplicities (3, 2, 1), three chiral generations, anomaly-cancelling hypercharges, the TBM zeroth-order pattern $\sin^2\theta_{12} = 1/3$.
- **Layer 1 — Structural form of corrections.** Properties of the *form* of the perturbative expansion that follow from cubic-group representation theory and uniqueness constraints. Hold for every bijection. Examples: the Cabibbo scale $\lambda^2 = 1/(2\pi^2)$ (Brillouin geometry, §2.2), the Wolfenstein $\mathrm{A}^2 = 2/3$ (off-democratic projection), $\sin^2\theta_{13} = \mathrm{A}^4\lambda^2$ (independent of A_2 at $\mathrm{O}(\lambda^2)$, as shown in §3.3 above).
- **Layer 2(a) — Operator-level structural.** Relations among Wilson coefficients of cubic-invariant substrate operators that hold for every bijection. Promoted from Layer 2(b) when the coefficient ratio reduces to a structural identity for the relevant representation.
- **Layer 2(a)-tractable — Sharply specified residual.** A coefficient ratio that reduces to a sharply specified structural quantity (with a representation-theoretic reading) not yet derived from substrate dynamics, but with concrete substrate calculations available to test it. Distinguished from Layer 2(b) by the sharpness of the residual question.
- **Layer 2(b) — Solution-specific.** Coefficient values are free parameters of the bijection.

The *diagnostic* for distinguishing 2(a)-tractable from generic 2(b) is whether the residual question reduces to a *sharply specified* numerical target — a single coefficient ratio with a representation-theoretic reading, testable by definite substrate calculations — rather than a generic direction in operator space. A 2(a)-tractable question may still have bijection-specific content; the label refers to the sharpness of the verification path, not to representation-theoretic forcing. The identity $\mathrm{A}^2 = (\mathrm{d}-1)/\mathrm{d} = \mathrm{C}_2(\mathrm{T}_1)/\mathrm{d}$ for $\mathrm{d} = 3$, by contrast, is forced by representation theory and makes $\sin^2\theta_{13} = \mathrm{A}^4\lambda^2$ Layer 1.

Layer status of Condition 2. The A_1 component is Layer 1: structurally protected against the A_1 perturbation by cubic geometry, no substrate input required. The A_2 component reduces to the sharply specified question — *why does $b_{23}/(b_{12}+b_{13})$ take exactly the value $4/3 = 2A^2$?* — concerning a single numerical ratio with the structural reading $4(\mathrm{d}-1)/(2\mathrm{d})$ for $\mathrm{d} = 3$. This was labeled *Layer 2(a)-tractable* — the residual question sharply specified, admitting definite outcomes from concrete staggered-fermion calculations on the cubic lattice — until the calculation was performed. We do not claim the answer is forced by representation theory alone: a direct substrate calculation in [18, §7.3] establishes that the natural cubic-symmetric Higgs-democratic spurion does not produce the 4/3 ratio (instead giving b_{12}:b_{13}:$b_{23} \approx 17$:1:1 in the heavy-quark hierarchy, off by a factor of ~24 from the required 4/3), and the relation requires substrate amplitudes with non-natural scaling δM_{ij}^sub $\propto$ m_{max(i,j)} plus a channel-specific enhancement factor $K_{23}/K_{12} \approx 2.8$. The substantive content is therefore bijection-specific — Layer 2(b) — and the verification path has since been executed: the routing computation of [18, §7.3] excludes the second-order route (the two-hop operator is a pure spin-taste monomial, orthogonal to every mass channel), leaving the recorded $O(\lambda^3)$ construction as the one unpursued analytical route. The 0.07σ empirical match at $\sin^2\theta_{12}$ confirms the bijection has the required alignment structure; the 7/6 sum rule remains the direct experimental test.

In standard effective field theory, the U-even and dominant U-odd coefficients $b_{12}+b_{13}$ and b_{23} would be independent Wilson coefficients. The structural relation $b_{23} = (4/3)(b_{12}+b_{13})$ is more naturally accommodated in frameworks where quantum mechanics is itself an emergent phenomenon arising from a deeper deterministic structure [15, 16, 17], since EFT coefficients can then be related by substrate-level mechanisms even absent a protecting EFT symmetry. The factorization $4/3 = 2A^2$ is consistent with this picture; whether it follows rigorously from the cubic-lattice realization of [18] is the open task referenced above.

Sharpened open task. In the AF basis (charged leptons diagonal at LO), second-order perturbative diagonalization of M_ℓ M_ℓ† generates contributions to the A_2 coefficients from cross-terms involving A_1. Direct computation shows that the framework's choice $A_1 = (\surd 2/3)(E_{12} - E_{13})$ produces a hierarchy-independent cross-term contribution b_{23}^cross $= -1/9$ in the m_e$^2 \to 0$ limit, while b_{12} and b_{13} receive no analogous cross-term contribution. Cond. 2 at the observable level then translates to a relation on the substrate's direct second-order corrections:

b_{23}^sub = (4/3)(b_{12}^sub + b_{13}^sub) + 1/9

The +1/9 augmentation is forced by A_1's perturbative structure; deriving the substrate corrections that satisfy this from the cubic-lattice Yukawa is the quantitative open task. The ratio $b_{23}^{\mathrm{sub}/(b_{12}}$sub+$b_{13}$^sub) at leading order in the hierarchy m_e/m_τ takes the value $4/3 = 4(d-1)/(2d)$, and the sharp version of the question is whether a one-loop staggered-fermion calculation on the cubic lattice produces this specific ratio from the $C_2(T_1)$ Casimir structure.

We treat Condition 2 as a substrate-level relation and quote our angle predictions on the understanding that the predictions sit at the layered level: their structural form is forced by cubic-group representation theory and the Layer 1 closure above, but the specific 4/3 coefficient is a Layer 2(b) substrate input: the sharply specified second-order computation has since been completed and excludes that derivation route ([18, §7.3]). Empirical agreement at 0.07σ on $\sin^2\theta_{12}$ is, in this sense, a nontrivial test of the substrate's specific operator structure.

At the angle level, Condition 2 is equivalent to

$\mathbf{2\Delta_{12} + \Delta_{23} = 0}$

which is testable directly against data (Sec. 4.3). The combination $b_{12} - b_{13}$ remains free; we identify it in Sec. 4.2 with δ_CP (the natural identification, since it is the only remaining $O(\lambda^2)$ free parameter that does not enter any of the three angles). The present analysis therefore predicts all three angles but not δ_CP.

3.5 Numerical predictions

Substituting $b_{23} = 1$, $b_{12} + b_{13} = 3/4$, and $\lambda^2 = 1/(2\pi^2)$:

- $\mathbf{\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.30798}$
- $\mathbf{\sin^2\theta_{13} = 4/(18\pi^2) = 0.02252}$
- $\mathbf{\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.55066}$

The first formula for $\sin^2\theta_{13}$ follows from $A^2 = 2/3$ and $\lambda^2 = 1/(2\pi^2)$ together with the constraints on A_1; it does not use the normalization conditions on A_2. The other two additionally use Conditions 1 and 2.

4. The $\sin^2\theta_{12}$ prediction

4.1 Numerical value and comparison with data

The headline prediction is

$$\mathbf{\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.30798}$$

Comparison with current measurements:

Source	$\sin^2\theta_{12}$	σ from prediction
This work	0.30798	—
JUNO direct	0.3092 ± 0.0087	0.14σ
Post-JUNO global	0.3085 ± 0.0073	0.07σ

The prediction agrees with the post-JUNO global fit at 0.07σ. Both inputs entering the prediction are independently motivated: 1/3 is the A_4 TBM zeroth-order value, and $\lambda^2 = 1/(2\pi^2)$ is the squared Cabibbo angle, fixed independently by the cubic Brillouin geometry (matching the observed $\lambda = 0.22508$ to 0.04%).

4.2 Comparison with column-preservation patterns

Tribimaximal mixing [1] was the leading phenomenological description of neutrino oscillation data until Daya Bay measured a nonzero θ_{13} [2]. Since then, modified TBM patterns have been the subject of a substantial literature [3, 4, 5, 6]. The two best-known modifications preserve one column of the original TBM matrix exactly: TM1 (first column) and TM2 (second column). Each gives a one-parameter family that accommodates the observed θ_{13} but predicts different values for the other two angles. Most existing TBM-modification proposals start from a discrete *flavor symmetry* — typically A_4, S_4, or a Z_2 residual — postulated as an axiom of the lepton sector. The minimal modifications then preserve one column of the TBM matrix or one residual symmetry of the neutrino mass matrix, yielding one-parameter correlations between angles. Post-JUNO, He [9] finds that TM2 is excluded at $>3.5\sigma$ while TM1 remains viable; Zhang [10] reports similar results.

Comparison summary:

Approach	Symmetry origin	$\sin^2\theta_{12}$	σ
TM1	postulated A_4 flavor	0.3184	1.35σ
TM2	postulated A_4 flavor	0.3408	4.4σ
TM3	postulated; $\theta_{13} = 0$	—	excluded
This work	**cubic group O of 3D space**	**0.3080**	$\mathbf{0.07\sigma}$

(Comparing against post-JUNO global fit 0.3085 ± 0.0073. TM1 uses $\sin^2\theta_{12} = 1 - (2/3)/\cos^2\theta_{13}$; TM2 uses $\sin^2\theta_{12} = 1/(3\cos^2\theta_{13})$; both with $\sin^2\theta_{13} = 0.02195$.)

Three differences distinguish the present prediction from TM_n:

(i) Symmetry origin. TM1, TM2, and TM3 take a discrete flavor symmetry as an axiom of the lepton sector. The cubic-group prediction here uses $A_4 \subset O$, where O is the symmetry group of three-dimensional space and the three generations transform as the T_1 irrep (Sec. 2.1; forced by the framework's Bravais-lattice uniqueness and anomaly-cancellation chain).

(ii) Independence of predictions. TM1 gives a one-parameter correlation, $\sin^2\theta_{12} = 1 - (2/3)/\cos^2\theta_{13}$, fixing θ_{12} given θ_{13} but predicting neither separately. The cubic-group analysis predicts all three angles as fixed numerical values, with no parameter common to them (Sec. 3.5).

(iii) CP violation. He's surviving pattern predicts sin δ_CP = ± 0.998, close to maximal. Here, δ_CP is identified with the only remaining free parameter at $O(\lambda^2)$, namely $b_{12} - b_{13}$, which does not enter any of the three angles. The angle predictions therefore stand independently of any value of δ_CP, and the present paper makes no prediction for it.

The bottom line: the cubic-group prediction is closer to the post-JUNO global fit by a factor of about 20 in σ-distance than the best surviving column-preservation pattern, and at the same time makes more falsifiable claims (three independent angles plus the sum rule below).

4.3 The sum-rule discriminator

The cubic-group prediction implies

$$\mathbf{2sin^2\theta_{12} + sin^2\theta_{23} = 7/6 = 1.1667}$$

versus the observed 2(0.3085) + 0.561 = 1.178 ± 0.020 (using the post-JUNO $\sin^2\theta_{12}$ and the NuFIT 6.0 NO best fit for $\sin^2\theta_{23}$). The agreement is at 0.6σ. This depends on the atmospheric octant: the quoted value uses the NuFIT 6.0 NO best fit $\sin^2\theta_{23} \approx 0.561$, the upper octant; under the lower-octant solution ($\sin^2\theta_{23} \approx 0.455$) the combination is ≈ 1.07, roughly 5σ below 7/6. This sum rule is the angle-level statement of $b_{23} = (4/3)(b_{12} + b_{13})$, which is structural in the cubic-group setup. TM1 and TM2 do not produce this combination as an exact sum rule; in TM2, $\sin^2\theta_{12}$ is determined by θ_{13} alone with no θ_{23} correlation. A direct measurement of $2\sin^2\theta_{12} + \sin^2\theta_{23}$ at the 0.005 level is therefore the sharpest data-side test that distinguishes the two classes of TBM modification.

5. The remaining angles

5.1 $\sin^2\theta_{13}$

The formula for $\sin^2\theta_{13}$ is fully derived: at $O(\lambda^2)$ the A_2 parameters cancel from |U_e3|^2, leaving $\sin^2\theta_{13} = A^4\lambda^2 = 4/(18\pi^2) = 0.02252$. The NuFIT 6.0 best fit (normal ordering) is 0.02195 ± 0.00056, agreeing at 1.01σ. Equivalently,

$$\mathbf{sin\theta_{13} = A^2\lambda = (2/3)\ \cdot\ 1/(\pi\sqrt{2}) = 0.1500}$$

giving $\theta_{13} = 8.63°$, compared with the observed 8.56° ± 0.10°.

5.2 $\sin^2\theta_{23}$

The formula gives $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.5507$. NuFIT 6.0 (NO) gives $\sin^2\theta_{23} = 0.561^{+0\,.\,012}{}_{-0\cdot 015}$, agreeing at 0.74σ. The prediction lies in the second octant, in line with the current global preference. The sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ ties this directly to $\sin^2\theta_{12}$:

$$\mathbf{sin^2\theta_{23} = 7/6 - 2sin^2\theta_{12}}$$

A ±0.0014 ultimate JUNO error on $\sin^2\theta_{12}$ then implies a ±0.0028 error on the predicted $\sin^2\theta_{23}$, comparable to the current direct measurement.

5.3 Summary

Angle	Prediction	Measurement	σ
$\sin^2\theta_{12}$	$1/3 - 1/(4\pi^2) = 0.3080$	0.3085 ± 0.0073	0.07
$\sin^2\theta_{13}$	$4/(18\pi^2) = 0.02252$	0.02195 ± 0.00056	1.01
$\sin^2\theta_{23}$	$1/2 + 1/(2\pi^2) = 0.5507$	$0.561^{+0\,.\,012}{}_{-0\cdot 015}$	0.74

All three predictions agree with observation within 1.1σ.

6. Robustness

6.1 Cumulative empirical match

The 0.07σ match for $\sin^2\theta_{12}$ does not stand alone. It sits within a set of empirical matches all using the same cubic-group structural inputs:

Observable	Structural input	Prediction	Observed	Match
Cabibbo angle	Brillouin distance $\pi\sqrt{2}$; chirality $1/\lvert q\rvert$	$\lambda = 1/(\pi\sqrt{2}) = 0.22508$	0.22500 ± 0.00067	0.04%
Koide angle (charged leptons)	C_2/d^2 for cubic group	$\theta_0 = 2/9 = 0.22222$	0.222204 ± 0.000010	0.02%
Solar mixing $\sin^2\theta_{12}$ (this work)	A_4 representation $1/3$ + Cond 2 cubic relation	$1/3 - 1/(4\pi^2) = 0.30798$	0.3085 ± 0.0073	0.07σ
PMNS sum rule (this work)	Cond 2 cubic relation	$2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$	1.178 ± 0.020	0.6σ
Reactor mixing $\sin^2\theta_{13}$ (this work)	A^2 λ projection geometry	$4/(18\pi^2) = 0.02252$	0.02195 ± 0.00056	1.01σ
Atmospheric mixing $\sin^2\theta_{23}$ (this work)	Cond 2 + sum rule	$1/2 + 1/(2\pi^2) = 0.55066$	$0.561^{+0.012}_{-0.015}$	0.74σ

Two structural parameters appear repeatedly: $\lambda^2 = 1/(2\pi^2)$, the squared Cabibbo angle from the cubic Brillouin geometry, and $A^2 = 2/3$, the off-democratic projection $\sin^2(\angle(\text{e\_i}, \hat{\text{h}}))$ from the cubic-group representation. Both enter multiple observables at sub-percent or sub-σ precision. Under any reasonable prior over flavor models, this is non-trivial structural agreement: a single set of inputs produces six distinct empirical matches, each of which would have to be explained as coincidence in a flavor-symmetry framework that postulates the cubic structure rather than deriving it.

6.2 Sensitivity to Condition 2

The factor 4/3 in Condition 2 (Sec. 3.4) is the one substantive structural input. It is *not* a free parameter fitted to $\sin^2\theta_{12}$: the identity $4/3 = 2\mathrm{A}^2$ with $\mathrm{A}^2 = (\mathrm{d}-1)/\mathrm{d} = \mathrm{C}_2(\mathrm{T}_1)/\mathrm{d}$ for d = 3 fixes 4/3 as a representation-theoretic quantity — the squared off-democratic projection of the vector representation of the cubic group, equivalently the quadratic Casimir of T_1 over the lattice dimension. Both $\mathrm{A}^2 = 2/3$ and the resulting 4/3 are determined by cubic-group representation theory independently of any measurement of θ_{12}. The same $\mathrm{A}^2 = 2/3$ enters the Cabibbo derivation through the Wolfenstein A (§2.2) and is independently fixed there by the cubic-group projection geometry. The 0.07σ match at $\sin^2\theta_{12}$ therefore tests a representation-theoretic identity, not a fitted coefficient.

The observable consequences of Condition 2 enter the sum rule $2\Delta_{12} + \Delta_{23} = 0$ and, through Conditions 1 and 2 jointly, the absolute predictions for $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$. A hypothetical shift to $4/3 \to (4/3)(1 + \varepsilon)$ propagates to $\Delta_{12} \to \Delta_{12} + (1/2) \cdot \varepsilon \cdot \Delta_{23}$ at first order in ε. With $\Delta_{23} \approx 1/(2\pi^2) \approx 0.05$, a fractional shift $\varepsilon = 0.10$ (a 10% shift in Cond 2's structural coefficient) gives a shift in $\sin^2\theta_{12}$ of ~0.0025 — comparable to JUNO's ultimate-precision uncertainty of ±0.0014 and roughly half of JUNO's first-result uncertainty of ±0.0087. The current 0.07σ agreement therefore tests Cond 2 at the few-percent level on its structural coefficient. The substantive content of this test is that 4/3 is precisely $2\mathrm{A}^2$ where $\mathrm{A}^2 = 2/3$ is the projection factor independently fixed by the cubic representation geometry — a falsifiable structural relation rather than a fitted parameter.

6.3 Status of Condition 2

Condition 2 admits the structural decomposition into two parts established in Sec. 3.4: an A_1 component closed at Layer 1 (the TBM sum rule is structurally protected against the A_1 perturbation by the $\cos^2\theta_{13}$ dressing alone, requiring no substrate input), and an A_2 component reducing to the sharply specified question why $\mathrm{b}_{23}/(\mathrm{b}_{12}+\mathrm{b}_{13})$ takes the specific value $4/3 = 2\mathrm{A}^2$ in the framework's bijection. The reactor angle $\sin^2\theta_{13}$ is independent of Condition 2 (the A_2 coefficients cancel from |U_e3|²) and remains Layer 1 unconditional structural. The angle predictions $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ are layered: their structural form is forced by cubic-group representation theory and the Layer 1 closure of the A_1 component, while the specific 4/3 coefficient is bijection-specific content (Layer 2(b) in strict classification) with a sharply specified verification path — a single numerical ratio with the structural reading $4(\mathrm{d}-1)/(2\mathrm{d})$ for d = 3 — since executed: the second-order route is excluded ([18, §7.3]).

The 4/3 coefficient: verification path executed. The verification path — a staggered-fermion calculation on the cubic lattice testing whether the two-hop Casimir structure produces the $\mathrm{b}_{23}/(\mathrm{b}_{12}+\mathrm{b}_{13}) = 4/3$ ratio — has been executed ([18, §7.3]): the two-hop operator is a pure spin-taste monomial, orthogonal to every mass channel, so the second-order route is excluded and the coefficient stands at Layer 2(b), with the recorded $O(\lambda^3)$ construction the one unpursued analytical route. The bijection space in [18] is constrained by linear dynamics, T-invariance, P-indivisibility, and cycle ergodicity, leaving a discretization-scheme freedom (field quantization, lattice size, timestep, boundary conditions, rounding rule) rather than abstract permutation freedom. The substrate calculation tests whether the bijection's specific structure produces the 4/3 coefficient required by Condition 2 — a single sharply defined numerical ratio whose value the framework's substrate-level analysis can compute. The empirical match at 0.07σ on $\sin^2\theta_{12}$ and 0.6σ on the sum rule provides experimental constraint that the bijection has the required alignment; the analytical question has since been closed by the staggered calculation ([18, §7.3]). This is the kind of verification path that distinguishes 2(a)-tractable content from intrinsically open-ended speculation: the question was sharp, the calculation was specified, and the answer was binary — and it was delivered: the coefficient is not produced by the second-order route, which is precisely the falsifiability the classification was designed to preserve.

Generic substrate considerations indicate the bijection has non-trivial structure. An explicit substrate calculation in [18, §7.3] establishes that the natural cubic-symmetric staggered Yukawa structure on T_1 BZ corners does not produce the 4/3 coefficient: both the leading and subleading channels produce equal off-diagonal $\mathrm{Y}^{(2)}$ entries by cubic symmetry, with the natural form yielding b_{12}:b_{13}:$\mathrm{b}_{23} \approx 17{:}1{:}1$ ($\mathrm{K}_{23}/\mathrm{K}_{12} \approx 2.8$ — off from 4/3 by ~24× at the ratio level). This is informative content rather than a failure: it tells us the framework's bijection has direction-dependent structure beyond the natural cubic-symmetric form, and the 0.07σ empirical match indicates this non-natural structure aligns with the 4/3 coefficient at the few-percent level. The executed verification path tested whether the bijection's specific amplitude structure — constrained by linear dynamics, T-invariance, P-indivisibility, and cycle ergodicity — produces this alignment through the second-order channel; the answer is negative ([18, §7.3]), so the empirical alignment, which stands, is not sourced by that channel.

Cubic-group consistency test. The PMNS sum rule via Condition 2 is one observable in a six-prediction simultaneous-match cluster derived from the structural identity $\mathrm{A}^2 = \mathrm{C}_2/\mathrm{d}$ on the d=3 cubic lattice. The other five are Wolfenstein A (1.2%), $\sin^2\theta_{13}$ (0.4%), Koide θ_0 (0.02%), m_u/m_d (0.6%), and |V_cb| (0.4σ). All six match observation at sub-percent or sub-σ precision simultaneously [18, §7.7]. The sum rule's 0.6σ match is therefore not an isolated coincidence but part of a broader pattern of cubic-group structural consistency across the framework: the same $\mathrm{A}^2 = 2/3$ projection factor and the same $\lambda^2 = 1/(2\pi^2)$ parameter appear in multiple observables at sub-percent precision.

Discriminating power of the sum rule. The sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ is the most directly testable consequence of Condition 2 and the prediction's principal discriminator against alternative flavor frameworks. The discriminating power of the sum rule against the TM1, TM2, and TM3 column-preservation patterns (Sec. 4.2, 4.3) is independent of the Layer 2(a) vs 2(b) status of the

4/3 coefficient: those patterns predict distinct sum rules that fail the data at 1.4σ–6σ, while the cubic-group sum rule matches at 0.6σ. The empirical match at the sum rule level is therefore independent confirmation of the framework's structural prediction against the principal competing frameworks, regardless of how the 4/3 coefficient verification path closes.

7. Falsifiability

The prediction is sharp: $\lambda^2 = 1/(2\pi^2)$ is fixed independently by the Cabibbo angle, and the coefficient $1/(4\pi^2)$ in Δ_{12} has no adjustable input. Three experimental directions test it.

Sub-percent precision on $\sin^2\theta_{12}$. Over its 30-year design lifetime, JUNO is projected to reach $\sin^2\theta_{12} = 0.3092 \pm 0.0014$, a $6\times$ improvement over the current first result. At that precision the prediction 0.30798 is tested at 0.86σ (using the JUNO 59-day central value) or 0.36σ (using the post-JUNO global central value). An ultimate JUNO 2σ disagreement with 0.30798 would falsify the prediction.

The sum rule. The prediction $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ is sharper than either angle alone: it ties JUNO's reactor measurement to the atmospheric measurements at DUNE, T2HK, and Hyper-Kamiokande. As both improve, the sum rule will be tested at the 0.005 level — a sharper test than either side separately.

δ_CP. The free parameter $\mathrm{b}_{12} - \mathrm{b}_{13}$ of A_2 does not enter any angle prediction, so the framework is consistent with any δ_CP. Long-baseline measurements at DUNE and Hyper-Kamiokande will determine this remaining parameter directly.

8. Conclusion

We have derived $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.3080$ from cubic-group flavor structure, matching the post-JUNO global fit at 0.07σ with no parameter fit to $\sin^2\theta_{12}$. The same construction predicts $\sin^2\theta_{13}$ and $\sin^2\theta_{23}$ within 1.1σ of the NuFIT 6.0 best fits [14], and forces the exact sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$, which distinguishes the prediction from the TM1, TM2, and TM3 column-preservation patterns. The reactor angle $\sin^2\theta_{13} = 4/(18\pi^2)$ is unconditional structural (Layer 1 per the classification of §3.4), based on the representation-theoretic identity $\mathrm{A}^2 = (\mathrm{d}-1)/\mathrm{d} = \mathrm{C}_2(\mathrm{T}_1)/\mathrm{d}$ for vector reps. The solar and atmospheric angle predictions are layered: their structural form is forced by cubic-group representation theory and the Layer 1 closure of the A_1 component (Sec. 3.4), while the specific 4/3 coefficient in Condition 2 ($\mathrm{b}_{23} = (4/3)(\mathrm{b}_{12}+\mathrm{b}_{13})$) is bijection-specific content — Layer 2(b), its sharply specified verification path since executed with a negative outcome for the second-order route ([18, §7.3]) — with the substrate ratio $\mathrm{b}_{23}{}^{\mathrm{sub}/(\mathrm{b}_{12}}$sub+$\mathrm{b}_{13}$^sub) constrained directly by the 7/6 sum rule. The empirical match at 0.07σ on $\sin^2\theta_{12}$ and at 0.6σ on the sum rule indicates the framework's bijection has the alignment property required by Condition 2; the predictions are therefore tests of the bijection's specific structure as much as of the underlying cubic-group framework. The Cabibbo scale $\lambda^2 = 1/(2\pi^2)$ entering the prediction is set independently by cubic Brillouin-zone geometry and matches the observed Cabibbo angle to 0.04%.

JUNO's projected long-term precision will test the prediction at sub-percent level over the next decade. Combined with atmospheric measurements at DUNE, T2HK, and Hyper-Kamiokande, this will sharply test the sum rule. A confirmed result at JUNO's ultimate precision would support cubic-group flavor structure as a viable origin of the lepton mixing pattern.

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